

AQUATIC - RAY

DESIGN CONCEPT -

AQUATICUS-RAY IS A BIOMIMETIC UNDERWATER RESEARCH HABITAT THAT DRAWS INSPIRATION FROM THE STINGRAY - A MASTER OF FLUID MOVEMENT, EFFICIENT FORM AND HARMONY WITH WATER

THE DESIGN AIMS TO CREATE A RESILIENT, SELF-SUSTAINING AND ADAPTIVE HABITAT THAT SUPPORTS RESEARCH, LEARNING AND HUMAN WELL-BEING WHILE MINIMIZING ENVIRONMENTAL IMPACT

★ ADAPTIVE BY NATURE

★ INTEGRATED WITH FLOW

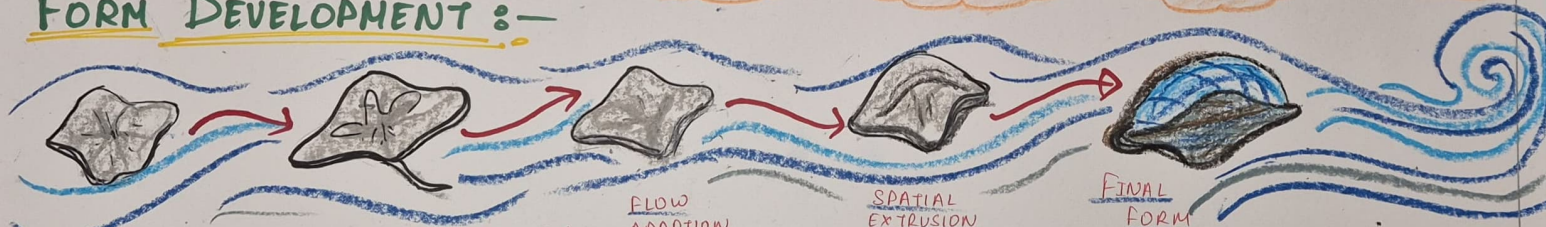
★ RESILIENT BY DESIGN

★ IN HARMONY WITH WATER

DESIGN INSPIRATION
THE STINGRAY



FORM DEVELOPMENT :-



NATURE STUDY

Study of stingray anatomy, morphology & movement dynamics.

FORM ABSTRACTION

Extraction key curves, central core and wings extensions.

FLOW ADAPTION

Refining form to reduce drag, optimize water flow & pressure distribution.

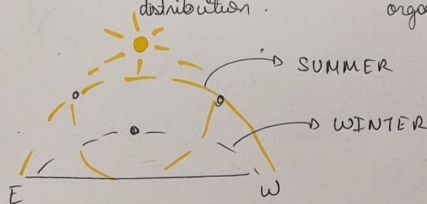
SPATIAL EXTRUSION

Translating the refined shell into 3D volumes and internal program organisation.

FINAL FORM

Integrated structural shell with functional systems for a complete underwater habitat.

SITE ANALYSIS :-



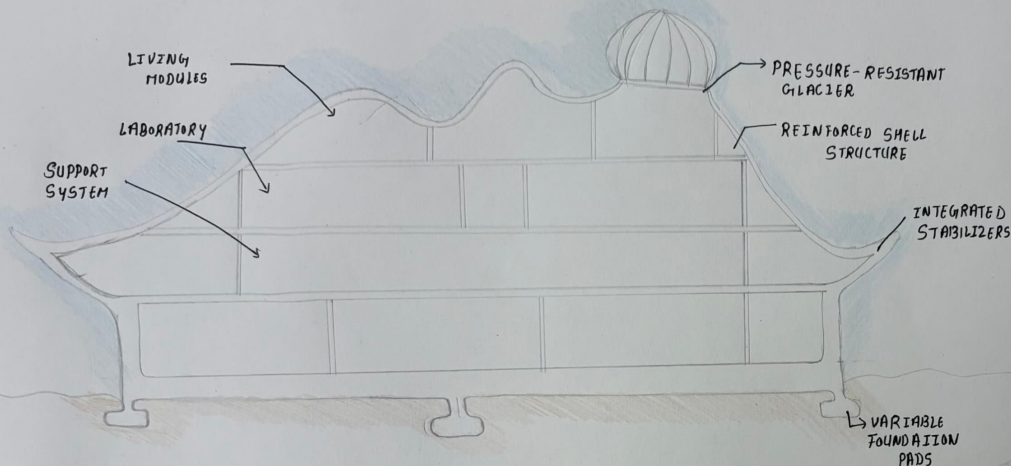
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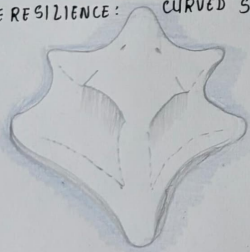
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SPATIAL LAYOUT | HYDRODYNAMIC PROFILE | MATERIALITY

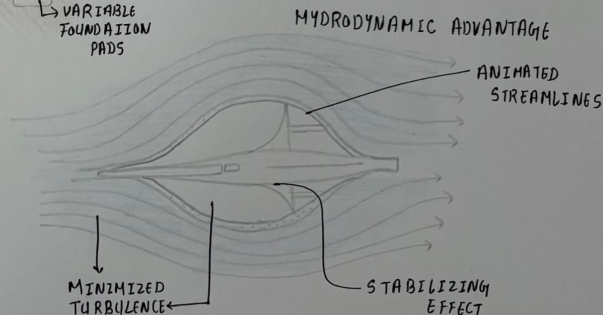


PRESSURE RESILIENCE: CURVED SURFACES MAXIMIZE COMPRESSIVE STRENGTH



CONCEPT: BIOMIMICRY OF THE STINGRAY

- The flat, low profile body of the stingray inspired the overall hydrodynamic form to minimize resistance to strong ocean currents
- The 'wings' shape maximizes the footprint on the ~~steady~~ sandy seabed for stability while creating extensive perimeter areas for a wide of living spaces.



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